



**UNIVERSITY OF COLOMBO, SRI LANKA**

*University of Colombo School of Computing*

**BACHELOR OF SCIENCE IN COMPUTER SCIENCE**

**Second Year Examination – Semester I – 2022**

**SCS2208 – Rapid Application Development** 177

**TWO (2) HOURS**



**To be completed by the candidate**

Examination Index No: .....

**Important Instructions to candidates:**

1. Students should answer in the medium of **English language only** using the space provided in this question paper.
2. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Write your index number **CLEARLY** on each and every page of this Question paper.
4. This **paper** consists of **04** questions in **14** pages (including the Cover Page).
5. Answer **ALL** questions.
6. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.
7. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

**For Examiner's use only**

| Question No | Marks |
|-------------|-------|
| 1           |       |
| 2           |       |
| 3           |       |
| 4           |       |
| Total       |       |

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**Question 1**

(a) What is the Rapid Application Development (RAD) Methodology?

[02 Marks]

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(b) List down the **four (04)** essential ingredients of the RAD.

[04 Marks]

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| ii.  |
| iii. |
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(c) Identify the key differences between the Joint Requirements Planning (JRP) Workshop and the Joint Application Design (JAD) Workshop.

[02 Marks]

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(d) List down **three (03)** major RAD roles.

[03 Marks]

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(e) Briefly explain the “**Time Box Management**” technique.

[04 Marks]

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(f) Briefly explain “**SWAT Team**” in the context of RAD.

[03 Marks]

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(g) List down **three (03)** benefits of RAD.

[03 Marks]

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| iii. |
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(h) What are the **four (04)** values of the Agile Manifesto?

[04 Marks]

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| i.   |
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| iii. |
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**Question 2**

Answer the following questions which are based on the fundamental concepts of JavaScript (JS).

(a) One of the most foundational principles that guide JavaScript is the preservation of “*backward compatibility*”. What is meant by backward compatibility?

[03 Marks]

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(b) Explain how a JS file will be executed in the JS Engine in a Browser.

[04 Marks]

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(c) Briefly explain the below given terms.

i. *Dynamic Typing*

[02 Marks]

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ii. *Coercion*

[02 Marks]

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iii. *Hoisting*

[02 Marks]

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iv. *Lexical Scoping*

[02 Marks]

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v. *Scope Chain*

[02 Marks]

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vi. *Events*

[02 Marks]

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vii. *Event listeners*

[02 Marks]

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(d) Write down the output of following scripts.

i.

```
var myArray = new Array(5);  
myArray[50000] = 3;  
console.log(myArray.length);
```

[02 Marks]

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ii.

```
var x;
console.log(typeof(x));
```

[02 Marks]

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**Question 3**(a) Briefly describe **synchronous** and **asynchronous** programming

[04 Marks]

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| i. <i>Synchronous programming</i>   |
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|                                     |
| ii. <i>Asynchronous programming</i> |
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(b) Consider the following code segment.

```
var x = "Hello";
var y = 10;
if(y>5){
    var x = "Hi";
}
console.log(x);
```

When executed it will print "Hi" in the console without raising an error. Briefly describe the reason for this and how to avoid this issue.

[ 04 Marks]

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- (c) Consider the following Mongosh commands used to create a collection database named "pos".

```
use pos
db.createCollection("users")
db.users.insertMany([{_id:1,name:"Amal",age:25,mark:75},{_id:2,name:"Kamal",age:27,mark:60},{_id:3,name:"Namal",age:25,mark:89},{_id:4,name:"Kamani",age:25,mark:94},{_id:5,name:"Ruvani",age:23,mark:55}])
```

Write down **Mongosh** commands for the following.

- i) Retrieve users whose age is greater than or equal to 25.

[01 Mark]

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- ii) Add 5 to the mark of all the users whose mark is less than 65.

[01 Mark]

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- iii) Delete all users whose age is 25.

[01 Mark]

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Write down the **output** of the following Mongosh command after successfully executing all the previous commands.

(iv) `db.users.find({}, {age:1})`

[02 Marks]

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(iv) `var cur =db.users.find()  
cur.sort({mark:-1})`

[02 Marks]

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(d)

(i) Consider the following information about a MongoDB collection.

| Field   | Value Description                   |
|---------|-------------------------------------|
| name    | May contain only letters            |
| email   | May contain alphanumeric characters |
| age     | An integer value                    |
| isAdmin | True or False                       |

Write down the **complete** content of the “user.js” file which will define and export a Mongoose Schema named “User”.

[05 Marks]

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ii) Briefly describe the functionality of the **body-parser** and **cors** modules.

[05 Marks]

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**Question 4**

(a) Briefly describe the **four (04)** components of the Selenium software suite.

[04 Marks]

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(b) Briefly describe **five (05)** agile business objectives.

[05 Marks]

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Index No: .....

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(c) Describe **three (03)** Content Management Systems types with their benefits.

[06 Marks]

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(d) Assume this is the start of your main.py file

```
from selenium import webdriver
from selenium.webdriver.firefox.service import Service as
FirefoxService
from webdriver_manager.firefox import GeckoDriverManager
from selenium.webdriver.common.by import By

service =
FirefoxService(executable_path=GeckoDriverManager().install())
driver = webdriver.Firefox(service=service)

driver.get("https://scs2208.ucsc.lk/")
```

Write the **Selenium code in Python** to achieve the following.

(i) Select and click on following link

`<a href="https://ucsc.lk">ucsc.lk</a>`

[02 Mark]

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(ii) Select and enter the email kav@ucsc.lk to the following input field.

```
<input type="email" class="form-control" id="uniEmail" aria-  
describedby="emailHelp" placeholder="Email">
```

[02 Mark]

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(iii) Check whether the following button is enabled and assign the result to variable isEnabled.

```
<button name="btnSubmit" type="submit">Submit</button>
```

[02 Mark]

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(iv) Retrieve size and position of the in the following heading and assign it to variable headingSize.

```
<h1>Hello. This is an example heading!</h1>
```

[02 Mark]

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(v) Retrieve the color of the following link and assign it to variable color.

```
<h2 style="color: red" >VLE</h2>
```

[02 Mark]

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